

Assignment 2

Assignment Instructions: Workshop Peer- Assessment

Assignment 2 is a **four-phase assessment**, and you must note and act according to the requirements for each phase. The instructions for each phase are as follows:

Phase 1 – The File Submission Phase

- **Complete the questions.** Write the code to answer the question.
- While developing the solution, write down the step-by-step instructions on how to solve the problem.
- The instructions should be provided below the code. This should be done in a Word document and converted to PDF for submission. **Both the code and instructions must be submitted in the PDF file.**
- You must save the PDF document with your student number and the assessment as the name e.g. 12345678_A2.pdf.

File Submission

- The assessment must be uploaded as a single Microsoft Word or PDF document.

The grading phase will open as soon as the submission phase has closed, you will have one (1) week to complete the grading phase.

Phase 2 - Testing

Once phase 1 (submission) is closed, you are required to return to the assessment. Download at least three (3) of your fellow student's step-by-step instructions, and follow the instructions provided to see if you can solve the programming problem.

The testing/grading phase will open as soon as the submission phase has closed, you will have one (1) week to complete the testing/grading phase.

Phase 3 - Grading

Return to the assessment for **phase 3 and grade the three (3) students' step-by-step instructions, as well as your own**, using the rubric that will be provided. You will thus **grade four (4) assessments in total**.

Your result will be dependent on how you grade yourself and your fellow students, as well as how you are graded. If you grade too high or too low in comparison with the other students who grade the same assessment, your grading will negatively impact your own grade. You thus need to make sure you provide a fair and honest grade.

The testing/grading phase will open as soon as the submission phase has closed, you will have one (1) week to complete the testing/grading phase.

Phase 4

Once the assessment phase is closed, return to your assessment. **Read the feedback and comments provided by your fellow students.** If you have any queries related to your feedback, please send an email to the lecturer detailing why you do not agree with the assessment you received. **You have one (1) week after the close of the grading phase to request moderation of your grade.**

Question 1

Explain what is wrong with the following code fragment and write code to correct it:

```
int main()
{
    PersonType family[20],
                newBaby("Anny Dube", 20180912, "2
    Sept");
    //Assume family has been initialised
    for (int i = 0; i < 20; i++)
        if (family.birthday[i] == newBaby.birthday)
            cout << family.name[i] << " share a
    birthday with "
                                <<newBaby.name;

    return 0;
}
```

Question 2

You have to write a program to read an input file character by character to help Peter solve the following activity in his activity book. The paragraph below is given:

We hope that you entered the activity. After you have completed the activity, the result is: The Activity Competition, Betty Davidson, 99, Auckland Park, 8989, and the chance to win a hamper consisting of 26 activities and 25 prizes.

Create an input file `activity.dat` with the paragraph above. The numbers 0 to 7 have to be replaced as follows:

- 0 must be replaced by s
- 1 must be replaced by g
- 2 must be replaced by o
- 3 must be replaced by y
- 4 must be replaced by v
- 5 must be replaced by n
- 6 must be replaced by f
- 7 must be replaced by j

Ask the user to input the names of the input and output files. Read the input file character by character, and write the character (if it stays the same) to the output file, or write the changed character to the output file. Call your output file `competition.txt`.

Allow the user to specify the names of the input and output files.